# Aashish Sheshadri

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## PROFILE

Twelve years building the layer that lets automated judgment reach production safely — currently the agentic SDLC platform; before that the company-wide standard governing how much autonomy an ML system is granted, and hierarchical reinforcement learning for cluster capacity. Individual contributor: sets technical direction across orgs with no reporting line. 5 peer-reviewed publications (362 citations, Google Scholar, Aug 2026), 1 issued patent and 1 pending application as lead inventor, and the AAAI HCOMP Test of Time Award for SQUARE — a benchmark for aggregating disagreeing human judgments into reliable evaluation signal, which is the problem that reappears today as annotation quality and model-graded evaluation.

## EXPERIENCE

**PayPal Inc, San Jose, CA** **June 2014 – Present**

*Machine Learning Architect (Sr. MTS)* Feb 2023 – Present

Set ML and infrastructure strategy across cloud platform, data, and developer productivity. Decide which problems get an ML solution vs. a heuristic, select modeling approaches, and govern production rollouts across ~1,550 applications.

**Agentic SDLC — Infra, Application, and Data**

- Built the multi-agent platform that carries an engineer's stated intent to production across three lifecycles: **Infra SDLC** (provisioning and environment management), **Application SDLC** (onboarding through per-change environments, functional-test gates, and canary promotion), and **Data SDLC**. Specialist agents behind a single orchestrator, sharing memory, policy, and a skill catalog. In production across all three lifecycles since August 2026; adoption and cycle-time numbers still being collected.
- Set the safety envelope, which is where the design effort went: no agent holds a write path, and every mutation is executed by deterministic code acting on an agent's proposal. Authority is budgeted by blast radius — an open-ended tool loop for the agent furthest from production, a short schema-validated pipeline for the one that touches it.
- Made promotion a code check rather than an instruction: an evidence-first evaluator scores acceptance criteria and a state transition keyed on that verdict holds the change in draft until it passes.

**ML systems and deployment governance**

- Designed the ML deployment framework — canary evaluation, shadow mode, **graduated autonomy** — now the company-wide production standard governing how much autonomy an ML system is granted. Introduced as a Staff engineer.
- Defined what evidence a system must produce before its autonomy widens, so the question is answered by a gate rather than a review meeting.
- Set autoscaling and capacity strategy, and built the hierarchical RL policy for cluster capacity allocation across ~1,550 applications (~100k pods). The hard part was reward function and curriculum engineering, not the policy architecture.
- Found and closed reward-specification failures under optimization pressure — cases where the policy optimized the written objective into an outcome I had not intended. Reward design, and the evaluation that catches where a reward diverges from its intent, is the part of this work that transfers.
- Ongoing collaboration with IBM Quantum on variational algorithms for risk modeling (published in *Quantum*, 2023).

*Staff Machine Learning Engineer (MTS 2)* Mar 2020 – Feb 2023

- Led ML-driven predictive autoscaling for ~1,500 applications, working toward five-nines availability while cutting compute spend ~25%.
- Built time-series demand forecasting models with confidence-calibrated predictions for ~1,500 applications, handling distribution shift across seasonal, promotional, and organic traffic patterns.
- Developed sequence classification models using BERT for real-time threat detection in information security — security-critical inference where false negatives are expensive.
- Collaborated with IBM Quantum on variational quantum algorithms for feature selection in credit risk modeling. Published in Quantum (2023).

*Senior Machine Learning Engineer (MTS 1)* Mar 2018 – Feb 2020

- Architected the enterprise ML platform: multi-tenant GPU scheduling with prioritized access across business units, remote kernel orchestration supporting PyTorch, H2O, R, and Ray workloads, and isolated environments from notebook experimentation through production model serving.
- Built pipeline orchestration layer: notebook scheduling via Airflow, automated training pipelines with managed retraining cadences, and DAG-based workflow composition.
- Built the parameterized-notebook core behind PPExtensions' `run_pipeline` magic (PayPal open-source, BSD-3 — github.com/paypal/PPExtensions): the command parser and the AST-level parameter substitution that type-checks and injects values into each notebook's parameter cell, which is what lets a pipeline hand state from one stage to the next. Also the CSV connector, logging and configuration layers.

*Senior Research Engineer* Mar 2016 – Feb 2018

- Co-authored the FASH64 and FASH256 specifications with Doug Crockford (JSON inventor) — fast non-cryptographic hash functions validated by statistical dispersion analysis and empirical collision-rate testing (crockford.com/fash64.pdf).
- Designed and open-sourced the **Seif Protocol** — an ECC-512 handshake over Node.js, MIT-licensed (github.com/paypal/seif-protocol) — protocol design, formal verification, and cryptographic systems engineering from first principles.
- Research in probabilistic analysis and randomness validation — mathematical foundations later applied to distribution modeling and uncertainty quantification in ML systems.

*Research Engineer* Jun 2014 – Feb 2016

- Built and open-sourced the cryptographic foundation of the Seif project under Doug Crockford, both MIT-licensed: **seifnode**, the Node.js native addon exposing ISAAC RNG, ECC-512, AES-GCM and SHA3-256 (**543 stars —** github.com/paypal/seifnode), and **seifrng**, the C++ entropy-mining and CSPRNG layer beneath it (github.com/paypal/seifrng). Presented at OSCON in 2015 and 2016.

**The University of Texas at Austin, Austin, TX** **Sept 2012 – May 2014**

**Graduate Research Assistant**

- Research in IR evaluation and human computation: aggregating noisy, disagreeing human judgments into reliable evaluation signal, and hybrid expert/non-expert judging for building test collections affordably — the evaluation-quality problems that recur today in annotation aggregation and model-graded evaluation.
- Built and open-sourced the SQUARE benchmark (MIT) — a common harness for comparing crowd-consensus algorithms (majority vote, GLAD, CUBAM, Dawid-Skene) under supervised, semi-supervised, and unsupervised regimes: github.com/utir/square, and a second-generation Java implementation adding n-fold evaluation and a worker-simulation framework at github.com/utir/square-2.0.

*Teaching Assistant* Sept 2012 – Dec 2012

- Programming Languages (undergraduate course)

**Carnegie Mellon University, Pittsburgh, PA** **Sept 2011 – May 2012**

**Research Associate**

- LiDAR-camera calibration for mapping and localization. Visual odometry and orthographic views for global localization of planetary rovers. Contributed to Astrobotic's Google Lunar XPRIZE effort.

## EDUCATION

*The University of Texas at Austin — M.S. Computer Science, GPA 3.90* Aug 2012 – May 2014

Computer Vision & IR Thesis Track

*Carnegie Mellon University — Audited Courses in Computer Vision, Robotics & Perception* Aug 2011 – May 2012

*PES Institute of Technology, Bangalore — B.S. Electronics & Communication, GPA 8.40, Honors* Jun 2007 – Aug 2011

## PUBLICATIONS

- Variational quantum algorithm for unconstrained black box binary optimization: Application to feature selection — *Quantum*, Jan 2023
- Mix and match: collaborative expert-crowd judging for building test collections accurately and affordably — *DESIRES*, Aug 2018
- A collaborative approach to IR evaluation (Masters Thesis) — *UT Digital Repository*, May 2014
- SQUARE: A Benchmark for Research on Computing Crowd Consensus (First Author, **Test of Time Award, Oct 2024**) — *AAAI HCOMP*, Nov 2013. Code: github.com/utir/square
- Position Estimation by Registration to Planetary Terrain (First Author) — *IEEE MFI*, Sept 2012
- Complementary Flyover and Rover Sensing for Superior Modeling of Planetary Features — *FSR*, July 2012

## PATENTS

- **Session Management System** (Lead Inventor) — US11818219, Issued Nov 14, 2023
- **Dynamic Autoscaling of Server Resources Using Intelligent Demand Analytic Systems** (Lead Inventor) — application US 17/826,071, filed May 26, 2022 (pending)

## TECHNICAL SKILLS

**Languages:** Python, C++, Java, JavaScript; Go, C, SQL
**Frameworks:** PyTorch, TensorFlow, Kubernetes, GCP/AWS/Azure, Spark, Ray, Jupyter
**Domains:** Agentic Systems & Tool-Use Harnesses, Reward & Curriculum Design, Evaluation Methodology, Reinforcement Learning, Deployment Safety & Oversight, ML Infrastructure, Time-Series Forecasting

## INVITED SPEAKER

QCon, Strata, MLConf, Deep Learning World, RE•WORK Enterprise AI Summit, OSCON, Fluent

## PEER REVIEWER

Information Processing & Management (IF 7.4), Artificial Intelligence (IF 5.1)